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Electronic apparatus enabling easy recognition of continuous shooting timing, method of controlling electronic apparatus, and storage medium

Abstract

An electronic apparatus that enables a user to easily recognize continuous shooting timing. The electronic apparatus controls to light an item during performing an exposure processing, accept an instruction for performing a continuous shooting, and control not to start to light the item when the exposure processing starts before a lapse of a predetermined lighting-off period from lighting off the item last time, while accepting the instruction for performing the continuous shooting.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of application Ser. No. 17/551,304, filed Dec. 15, 2021, the entire disclosure of which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to an electronic apparatus that enables easy recognition of continuous shooting timing, a method of controlling the electronic apparatus, and a storage medium.
Description of the Related Art
(2) Shooting using an electronic shutter produces no shutter sound since a mechanical shutter is not driven, and hence it is sometimes impossible to recognize the shooting timing. Japanese Laid-Open Patent Publication (Kokai) No. H11-88824 discloses a technique for expressing, when shooting is being performed in a continuous shooting mode, a continuous shooting speed by blinking the display indicating the continuous shooting mode at a period set according to the continuous shooting speed.
(3) However, the technique disclosed in Japanese Laid-Open Patent Publication (Kokai) No. H11-88824 has a problem that a user cannot recognize the continuous shooting timing.
SUMMARY OF THE INVENTION
(4) The present invention provides an electronic apparatus that enables a user to easily recognize continuous shooting timing, a method of controlling the electronic apparatus, and a storage medium.
(5) In a first aspect of the present invention, there is provided an electronic apparatus, including a processor; and a memory storing a program which, when executed by the processor, causes the electronic apparatus to: control to light an item during performing an exposure processing, accept an instruction for performing a continuous shooting, and control not to start to light the item when the exposure processing starts before a lapse of a predetermined lighting-off period from lighting off the item last time, while accepting the instruction for performing the continuous shooting.
(6) In a second aspect of the present invention, there is provided a method of controlling an electronic apparatus, including controlling to light an item during performing an exposure processing, accepting an instruction for performing a continuous shooting, and controlling not to start to light the item when the exposure processing starts before a lapse of a predetermined lighting-off period from lighting off the item last time, while accepting the instruction for performing the continuous shooting.
(7) According to the present invention, a user is enabled to easily recognize the continuous shooting timing.
(8) Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1**A and **1**B are views of the appearance of a digital camera as an electronic apparatus according to an embodiment of the present invention.
(2) FIG. **2** is a block diagram showing an example of the configuration of the digital camera.
(3) FIG. **3** is a flowchart of a shooting mode process performed by the digital camera.
(4) FIGS. **4**A to **4**F are views each showing an example of a shooting live view displayed on a

display section.
(5) FIG. **5** is view showing an example of a menu setting screen displayed on the display section.
(6) FIG. **6**A is a flowchart of a shooting process.
(7) FIG. **6**B is a continuation of FIG. **6**A.
(8) FIGS. **7**A to **7**E are diagrams useful in explaining states of a white frame item which is lighted-on and lighted-off by the shooting process.
DESCRIPTION OF THE EMBODIMENTS
(9) The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof. Although in the present embodiment, the description will be given of a case where the present invention is applied to a digital camera as an electronic apparatus, the present invention is not limitatively applied to the digital camera. For example, the present invention may be applied to an apparatus equipped with a continuous shooting function for performing continuous shooting based on a shutter speed and a frame speed that are set by a user to selected ones of a plurality of choices.
(10) FIGS. **1**A and **1**B are views of the appearance of a digital camera **100** as the electronic apparatus according to an embodiment of the present invention. The digital camera **100** is capable of operating in a drive mode (continuous shooting mode) in which continuous shooting is executed while an operation member for instructing the start of shooting is being pressed. The number of images to be continuously shot per unit time in the drive mode can be set by a user. The number of images to be continuously shot per unit time in the drive mode is referred to as the frame speed or the continuous shooting speed. In the present embodiment, as the frame speed, one of 3 frames (low speed), 10 frames (normal speed), and 30 frames (high speed) can be selected. Note that the frame speed may be selectable from more candidates, or may be continuously settable by using a slider or the like.
(11) Further, in the digital camera **100**, an exposure time can be set as a shooting parameter. The exposure time is expressed as a shutter speed (Tv). As the shutter speed, one of a low-speed setting, such as ⅕ seconds, 1/30 seconds, and 1/60 seconds, a high-speed setting, such as 1/2000 seconds and 1/8000 seconds, and a long shutter, such as 10 seconds, can be set. Further, it is also possible to automatically set the shutter speed (perform the automatic exposure (AE) processing) according to a brightness of a subject for shooting, which is acquired in a shooting preparation state.
(12) The digital camera **100** displays an item for use in causing a user to recognize the shooting timing while continuous shooting is being executed in the drive mode. Further, the digital camera **100** has a characteristic configuration that the digital camera **100** displays the above-mentioned item at the timing of execution of shooting. This enables a user to easily recognize the timing of execution of shooting during continuous shooting.
(13) The digital camera **100** displays the item at the timing of execution of shooting for a time period associated with the exposure time. This enables a user not only to easily recognize the timing of execution of shooting during continuous shooting, but also to easily recognize the length of the exposure time.
(14) In a case where the exposure time is shorter than a predetermined threshold value, the digital camera **100** displays the item for a predetermined lighting-on period (minimum lighting-on period) regardless of the exposure time. This enables a user to easily recognize the timing of execution of shooting during continuous shooting even when the exposure time is set to be short (the shutter speed is set to a high speed).
(15) Further, the digital camera **100** controls the display such that a next display of the item is not started at least before a predetermined lighting-off period (minimum lighting-off period) elapses after the display of the item is completed. This makes it possible to prevent a user from erroneously recognizing that the exposure time is longer than the set exposure time due to a too short display interval of the item.
(16) In the following, the control of the digital camera **100** will be described in detail.

(17) FIG. **1**A is a front perspective view of the digital camera **100**. FIG. **1**B is a rear perspective view of the digital camera **100**.
(18) A display section **28** is provided on the rear side of the digital camera **100** and displays an image and various information. A touch panel **70***a* can detect a touch operation performed on a display surface of the display section **28**. An ex-finder display section **43** is provided on the top side of the digital camera **100** and displays various settings of the digital camera **100**, such as a shutter speed and an aperture value. A shutter button **61** is an operation member for giving a shooting instruction (image capturing instruction). A mode switching switch **60** is an operation member for switching between various modes. Terminal covers **40** are covers for protecting connectors (not shown) for connecting the digital camera **100** to external devices.
(19) A main electronic dial **71** is a rotation operation member, and settings, such as a shutter speed and an aperture value, can be changed by rotating the main electronic dial **71**. A power switch **72** is an operation member for switching between on and off of power supply of the digital camera **100**. A sub electronic dial **73** is a rotation operation member, and by rotating the sub electronic dial **73**, it is possible to perform operations including movement of a selection frame (cursor), feed of an image, and so forth. A four-direction key **74** is configured such that upper, lower, left, and right portions thereof can be pushed in, to thereby causing execution of processing associated with the pushed portion of the four-direction key **74**. A SET button **75** is a push button and is mainly used e.g. for determining selection of an item. A multi-controller (hereinafter referred to as the MC) **65** can receive direction-instructing operations for eight directions and an operation of pushing in a central portion thereof.
(20) A moving image button **76** is used for instructing the start and stop of moving image shooting (recording). An AE lock button **77** is a push button, and the exposure state can be fixed by pressing the AE lock button **77** in a shooting standby state. An enlarging button **78** is an operation button for switching on and off of an enlargement mode on the live view display (LV display) in a shooting mode. When the main electronic dial **71** is operated after the enlargement mode is set to on, the live view image (LV image) can be enlarged or reduced. In a reproduction mode, the enlarging button **78** functions as an operation button for enlarging a reproduced image or increasing the magnification rate of the image. A reproduction button **79** is an operation button for switching between the shooting mode and the reproduction mode. When the reproduction button **79** is pressed during the shooting mode, it is possible to shift the shooting mode to the reproduction mode and display a latest image out of images recorded in a recording medium **200** (referred to hereinafter), on the display section **28**. A menu button **81** is a push button used for instructing the display of a menu screen, and when the menu button **81** is pressed, the menu screen on which various settings can be made is displayed on the display section **28**. A user can intuitively make various settings, using the menu screen displayed on the display section **28**, and by operating the four-direction key **74**, the SET button **75**, or the MC **65**. A line-of-sight determination button **82** is an operation member included in an operation section **70** (referred to hereinafter) and is a push button for instructing selection or deselection of a subject for shooting based on a position of a user's line of sight within a viewfinder screen, which is detected by a line-of-sight detection sensor **164**, referred to hereinafter. The line-of-sight determination button **82** is disposed at a location easily operable by a user even in a state in which the user looks in a viewfinder (state in which a user's eye is in contact with an eyepiece **16**), which is a location operable by the user with the thumb of the user's right hand holding a grip portion **90**.
(21) A communication terminal **10** is used by the digital camera **100** so as to communication with a lens unit **150** (described hereinafter, and removable). The eyepiece **16** is a portion of an eyepiece viewfinder (look-in type viewfinder), and a user can visually recognize a video displayed on an internal electronic viewfinder (EVF) **29** through the eyepiece **16**. Note that the eyepiece viewfinder is formed by the eyepiece **16**, a line-of-sight detection section **160**, described hereinafter, and the EVF **29**. An eye-contact detection section **57** is an eye-contact detection sensor for detecting

whether or not an eye of a user (photographer) is in contact. A cover **202** is a cover for a slot that accommodates the recording medium **200** (described hereinafter). The grip portion **90** is a holding portion shaped such that the grip portion **90** can be easily grasped with a right hand when a user holds the digital camera **100**. The shutter button **61** and the main electronic dial **71** are arranged at respective locations operable with the forefinger of the right hand in a state holding the digital camera **100** by grasping the grip portion **90** with the little, ring, and middle fingers of the right hand. Further, the sub electronic dial **73** and the line-of-sight determination button **82** are arranged at respective locations operable with the thumb of the right hand in the same state.

(22) FIG. **2** is a block diagram showing an example of the configuration of the digital camera **100** shown in FIGS. **1**A and **1**B. The lens unit **150** is a lens unit on which an interchangeable photographic lens is mounted. Although a lens **103** is generally formed by a plurality of lenses, in FIG. **2**, the lens **103** is illustrated as only one lens for simplification. A communication terminal **6** is used by the lens unit **150** so as to communication with the digital camera **100**, and the communication terminal **10** is used by the digital camera **100** so as to communication with the lens unit **150**. The lens unit **150** communicates with a system controller **50** via these communication terminals **6** and **10**. Further, the lens unit **150** includes a lens system control circuit **4** that controls a diaphragm **1** via a diaphragm drive circuit **2**. Further, the lens system control circuit **4** of the lens unit **150** causes displacement of the lens **103** via an AF drive circuit **3** to thereby perform focusing.

(23) A shutter **101** is a focal plane shutter capable of freely controlling the exposure time of an image capturing section **22** under the control of the system controller **50**.

(24) The image capturing section **22** is an image capturing device implemented by a CCD or CMOS device that converts an optical image to electrical signals. The image capturing section **22** may have an imaging plane phase difference sensor that outputs defocus amount information to the system controller **50**.

(25) An image processor **24** performs predetermined processing (pixel interpolation, resizing, such as size reduction, color conversion, and so forth) on data output from an analog-to-digital converter **23** or data output from a memory controller **15**. Further, the image processor **24** performs predetermined computation processing on captured image data, and the system controller **50** performs exposure control and ranging control based on results of the computation by the image processor **24**. With these controls, auto-focus (AF) processing of a trough-the-lens (TTL) method, automatic exposure (AE) processing, electronic flash pre-emission (EF) processing, and so forth are performed. The image processor **24** further performs predetermined computation processing on captured image data, and performs automatic white balance (AWB) processing of the TTL method, based on obtained results of the computation.

(26) The memory controller **15** controls transmission/reception of data between the analog-to-digital converter **23**, the image processor **24**, and a memory **32**. Data output from the analog-to-digital converter **23** is written into the memory **32** via the image processor **24** and the memory controller **15**. Alternatively, data output from the analog-to-digital converter **23** is written into the memory **32** via the memory controller **15** without via the image processor **24**. The memory **32** stores image data captured by the image capturing section **22** and converted to digital data by the analog-to-digital converter **23** and image data to be displayed on the display section **28** or the EVF **29**. The memory **32** has a sufficient storage capacity to store a predetermined number of still images and a predetermined duration of moving image and sound.

(27) Further, the memory **32** also functions as a memory for image display (video memory). Image data for display written in the memory **32** is displayed on the display section **28** or the EVF **29** via the memory controller **15**. The display section **28** and the EVF **29** each perform the display on a display device, such as an LCD or an organic EL, according to signals delivered from the memory controller **15**. Data converted from analog to digital by the analog-to-digital converter **23** and accumulated in the memory **32** is sequentially transferred to the display section **28** or the EVF **29** so as to be displayed thereon, whereby a live view display (LV) is performed. An image displayed

by the live view display is hereinafter referred to as the live view image (LV image).
(28) The line-of-sight detection section **160** detects a line of sight of a user's eye which is in contact with the eyepiece **16** and views the EVF **29**. The line-of-sight detection section **160** is formed by a dichroic mirror **162**, an image forming lens **163**, the line-of-sight detection sensor **164**, a line-of-sight detection circuit **165**, and an infrared light-emitting diode **166**.
(29) The infrared light-emitting diode **166** is a light emitting element used for detecting the position of a user's line of sight within the viewfinder screen and irradiates an eyeball (eye) **161** of a user with infrared light. The infrared light emitted from the infrared light-emitting diode **166** is reflected by the eyeball (eye) **161**, and the reflected infrared light reaches the dichroic mirror **162**. The dichroic mirror **162** reflects only infrared light and allows visible light to pass therethrough. The reflected infrared light whose optical path has been changed forms an image on an imaging surface of the line-of-sight detection sensor **164** via the image forming lens **163**. The image forming lens **163** is an optical member which is a component of a line-of-sight detection optical system. The line-of-sight detection sensor **164** is implemented by an image capturing device, such as a CCD image sensor.
(30) The line-of-sight detection sensor **164** photoelectrically convers the incident reflected infrared light to electrical signals and outputs the electrical signals to the line-of-sight detection circuit **165**. The line-of-sight detection circuit **165** detects a position of the line of sight of the user from movement of the user's eyeball (eye) **161** based on the signals output from the line-of-sight detection sensor **164** and outputs the detection information to the system controller **50** and a gaze determination section **170**.
(31) Based on detection information received from the line-of-sight detection circuit **165**, the gaze determination section **170** determines, in a case where a time period in which the user's line of sight is fixed to one area exceeds a predetermined threshold value, that the user gazes the one area. Therefore, it can be said that the one area is a gazed position (gazed area) where the user has gazed. Note that the state in which “the line of sight is fixed to one area” refers e.g. to a state in which, until the lapse of a predetermined time period, an averaged position of movement of the line of sight is within the one area and the movement has variation (dispersion) less than a predetermined value. Note that the predetermined threshold value can be changed as desired by the system controller **50**. Further, instead of providing the gaze determination section **170** as an independent block, the system controller **50** may execute the same function as the gaze determination section **170** based on the detection information received from the line-of-sight detection circuit **165**.
(32) In the present embodiment, the line-of-sight detection section **160** detects a line of sight by a method called the corneal reflection method. The corneal reflection method is a method of detecting a direction and a position of the line of sight, based on a positional relationship between the reflected light of infrared light emitted from the infrared light-emitting diode **166** and reflected by the eyeball (eye) **161** (particularly, a cornea), and the pupil of the eyeball (eye) **161**. Note that the method of detecting the line of sight (the direction and position of the line of sight) is not particularly limited, but any other suitable method may be employed. For example, a method called the sclera reflection method may be used which makes use of a difference in light reflectance between a black portion and a white portion of an eye.
(33) The ex-finder display section **43** displays various settings of the camera, such as a shutter speed and an aperture value, via an ex-finder display section drive circuit **44**.
(34) A nonvolatile memory **56** is an electrically erasable and recordable memory and is implemented e.g. by a Flash-ROM. The nonvolatile memory **56** records constants, programs, and so forth, for the operation of the system controller **50**. The programs mentioned here refer to programs for executing various processes, described hereinafter, in the present embodiment.
(35) The system controller **50** is a controller implemented by at least one processor or circuit and controls the overall operation of the digital camera **100**. The system controller **50** realizes processes of the present embodiment, described hereinafter, by executing the programs recorded in the above-

mentioned nonvolatile memory **56**. A system memory **52** is e.g. a RAM, and the system controller **50** loads the constants, the variables, and the programs read from the nonvolatile memory **56** into the system memory **52**, for the operation of the system controller **50**. Further, the system controller **50** performs display control by controlling the memory **32**, the display section **28**, and so forth.
(36) A system timer **53** is a time measurement section for measuring a time used for various controls and a time of a built-in clock.
(37) A power supply controller **80** is formed by a battery detection circuit, a DC-DC converter, a switch circuit for switching blocks to be energized, and so forth, and detects whether or not a battery is attached, a type of the battery, a remaining charge of the battery, and so forth. Further, the power supply controller **80** controls the DC-DC converter based on results of the above-mentioned detection and an instruction from the system controller **50** to supply a required voltage to each of components including the recording medium **200** for a required time period. A power supply section **30** is comprised of a primary battery, such as an alkaline battery or a lithium battery, or a secondary battery, such as a NiCd battery, a NiMH battery, or an Li battery, and an AC adapter.
(38) A recording medium interface **18** is an interface with the recording medium **200**, such as a memory card or a hard disk. The recording medium **200** is a recording medium, such as a memory card, for recording a shot image, and is implemented by a semiconductor memory, a magnetic disk, or the like.
(39) A communication section **54** transmits and receives a video signal and an audio signal to and from an external device connected wirelessly or by a wired cable. The communication section **54** can be connected to a wireless Local Area Network (LAN) and the Internet. Further, the communication section **54** can communicate with an external device using Bluetooth (registered trademark) or Bluetooth Low Energy. The communication section **54** is capable of transmitting an image shot by the image capturing section **22** (including an LV image) and an image recorded in the recording medium **200** and receiving image data and other various information from an external device.
(40) A posture detection section **55** detects a posture of the digital camera **100** with respect to the gravity direction. Whether an image shot by the image capturing section **22** is an image shot when the digital camera **100** is held horizontally or an image shot when the digital camera **100** is held vertically can be determined based on a posture detected by the posture detection section **55**. The system controller **50** can add direction information dependent on a posture detected by the posture detection section **55** to an image file of an image captured by the image capturing section **22** and record an image after rotating the same. As the posture detection section **55**, an acceleration sensor, a gyro sensor, or the like can be used. It is also possible to detect movement of the digital camera **100** (whether the digital camera **100** is panned, tilted, lifted, or still) using the acceleration sensor or the gyro sensor as the posture detection section **55**.
(41) The eye-contact detection section **57** is an eye-contact detection sensor that detects approach of the eye (object) **161** to the eyepiece **16** of the eyepiece finder (hereinafter simply referred to as the “viewfinder”) (eye-contacted state) and separation of the eye (object) **161** from the eyepiece **16** (eye-removed state) (approach detection). The system controller **50** switches display (display state)/non-display (non-display state) of the display section **28** and the EVF **29** according to a state detected by the eye-contact detection section **57**. More specifically, in a case where the digital camera **100** is at least in the shooting standby state and the switching of a display destination is set to automatic switching, when a user's eye is not in contact with the eyepiece **16**, the display destination is set to the display section **28** and the display thereof is turned on, whereas the EVF **29** is set to the non-display state. On the other hand, when a user's eye is in contact with the eyepiece **16**, the display destination is set to the EVF **29** and the display thereof is turned on, whereas the display section **28** is set to the non-display state. As the eye-contact detection section **57**, an infrared proximity sensor, for example, can be used, and the infrared proximity sensor can detect proximity of any object to the eyepiece **16** of the viewfinder incorporating the EVF **29**. If an object

approaches, infrared light emitted from a light projection section (not shown) of the eye-contact detection section **57** is reflected by the object and received by a light reception section (not shown) of the infrared proximity sensor. A distance from the approaching object to the eyepiece **16** (eye approaching distance) can also be determined based on an amount of received infrared light. Thus, the eye-contact detection section **57** performs eye-contact detection for detecting an approaching distance from an object to the eyepiece **16**. Note that in a case where an object approaching within a predetermined distance from the eyepiece **16** is detected from the non-eye-contacted state (non-proximity state), it is determined that the object is in contact with the eyepiece **16**. In a case where an object which has been detected as the approaching object moves more than a predetermined distance away from the eye-contacted state (proximity state), it is determined that the object has been removed from the eyepiece **16**. The threshold value for detecting contact of an eye and the threshold value for detecting removal of an eye may be made different e.g. by setting a hysteresis. Further, after contact of an eye is detected, it is assumed that the eye-contacted state continues until removal of the eye is detected. After removal of the eye is detected, it is assumed that the non-eye-contacted state continues until contact of an eye is detected. Note that the infrared proximity sensor is described by way of example, but any other suitable sensor may be employed for the eye-contact detection section **57** insofar as it is a sensor capable of detecting proximity of an eye or an object, which can be regarded as eye-contacted.

(42) The system controller **50** can detect the following states of a line of sight with respect to the EVF **29** by controlling the line-of-sight detection section **160**: a state in which a line of sight which has not been directed to the EVF **29** is newly directed to the EVF **29**, i.e. the start of an input of the line of sight; a state in which the line of sight is being input to the EVF **29**; a state in which the user is gazing at one position in the EVF **29**; a state in which the line of sight directed to the EVF **29** is removed, i.e. the end of the input of the line of sight; and a state in which no line of sight is being input to the EVF **29** (state in which the user does not view the EVF **29**).

(43) These operations/states and the position (direction) of the line of sight directed to the EVF **29** are notified to the system controller **50** via an internal bus, and the system controller **50** determines how a line of sight is being input based on the notified information.

(44) The operation section **70** is an input section that receives an operation from a user (user operation) and is used to input various operation instructions to the system controller **50**. As shown in FIG. **2**, the operation section **70** includes the mode switching switch **60**, the shutter button **61**, the power switch **72**, and the touch panel **70***a*. Further, the operation section **70** includes the main electronic dial **71**, the sub electronic dial **73**, the four-direction key **74**, the SET button **75**, the moving image button **76**, the AE lock button **77**, the enlarging button **78**, the reproduction button **79**, the menu button **81**, the MC **65**, and so forth.

(45) The mode switching switch **60** is used to switch the operation mode of the system controller **50** to one of a still image-shooting mode, a moving image-shooting mode, the reproduction mode, and so forth. Examples of modes included in the still image shooting mode include an auto shooting mode, an auto scene determination mode, a manual mode, an aperture priority mode (Av mode), a shutter speed priority mode (Tv mode), and a program AE mode (P mode). Further, there are various scene modes in which different shooting settings are set for respective shooting scenes, a custom mode, and so forth. A user can directly switch the mode to one of these modes by using the mode switching switch **60**. Alternatively, after once switching the screen to a screen of a shooting mode list by using the mode switching switch **60**, the user may selectively switch the mode to one of the plurality of modes displayed on the list by using another operation member. Similarly, a plurality of modes may be included in the moving image-shooting mode.

(46) The shutter button **61** includes a first shutter switch **62** and a second shutter switch **64**. The first shutter switch **62** is turned on by a halfway operation, i.e. so-called half depression (shooting preparation instruction) of the shutter button **61** to generate a first shutter switch signal SW**1**. In response to the first shutter switch signal SW**1**, the system controller **50** starts shooting preparation

operations, such as the auto focus (AF) processing, the auto exposure (AE) processing, the auto white balance (AWB) processing, and the flash preliminary light emission (EF) processing. The second shutter switch **64** is turned on by a complete operation, i.e. so-called full depression (shooting instruction) of the shutter button **61** to generate a second shutter switch signal SW**2**. In response to the second shutter switch signal SW**2**, the system controller **50** starts a sequence of shooting processing operations from reading of signals from the image capturing section **22** to writing of a shot image into the recording medium **200** as an image file.

(47) The touch panel **70***a* and the display section **28** can be integrally formed. For example, the touch panel **70***a* is configured such that the light transmittance thereof does not block the display section **28** from displaying and is attached to the top layer of the display surface of the display section **28**. Further, input coordinates on the touch panel **70***a* and display coordinates on the display surface of the display section **28** are associated with each other. With this, it is possible to provide a graphical user interface (GUI) enabling a user to perform an operation as if the user directly operates the screen displayed on the display section **28**.

(48) The system controller **50** can detect the following operations performed on the touch panel **70***a* and the following states of the same: an operation of newly touching the touch panel **70***a* with a finger or a pen which has not been touching the touch panel **70***a*, i.e. the start of a touch (hereinafter referred to as a touch-down); a state in which the touch panel **70***a* is being touched with the finger or the pen (hereinafter referred to as a touch-on); a state in which the finger or the pen is moving while touching the touch panel **70***a* (hereinafter referred to as a touch-move); an operation of removing (releasing) the finger or the pen which has been touching the touch panel **70***a*, i.e. the end of a touch (hereinafter referred to as a touch-up); and a state in which nothing is touching the touch panel **70***a* (hereinafter referred to as a touch-off).

(49) When a touch-down is detected, a touch-on is detected at the same time. After the touch-down is detected, usually, the touch-on is continuously detected unless a touch-up is detected. In a case where a touch-move is detected, the touch-on is also detected at the same time. Even when the touch-on is detected, unless the touched position is moved, a touch-move is not detected. When a touch-up of all fingers or a pen which have/has been touching the touch panel **70***a* is detected and thereafter, the touch panel **70***a* is in the state of the touch-off.

(50) A detected one of operations and states and coordinates of a position on the touch panel **70***a* where a finger or a pen is touching are notified to the system controller **50** via the internal bus. Then, the system controller **50** determines what operation (touch operation) has been performed on the touch panel **70***a* based on the notified information. When a touch-move is detected, a moving direction in which the finger or the pen is moving on the touch panel **70***a* can also be determined with respect to each of vertical and horizontal components of the movement on the touch panel **70***a* based on changes of the position coordinates. In a case where a touch-move over a predetermined distance or more is detected, it is determined that a slide operation has been performed. An operation of quickly moving the finger or the pen over a certain distance while keeping the finger or the pen in contact with the touch panel **70***a* and then directly moving the finger or the pen off the same is referred to as a flick. In other words, the flick is an operation of quickly moving a finger or a pen on the touch panel **70***a* as if flicking the finger or the pen against the touch panel **70***a*. When a touch-move at a predetermined speed or higher over a predetermined distance or more has been detected and then directly a touch-up is detected, it is possible to determine that a flick has been performed (it is possible to determine that a flick has been performed after a slide operation). Further, a touch operation of touching a plurality of points (e.g. two points) at the same time (multi-touch) and moving the touched positions closer to each other is referred to as a pinch-in, and a touch operation of moving the touched positions away from each other is referred to as a pinch-out. The pinch-out and the pinch-in are collectively referred to as the pinch operation (or simply referred to as the pinch). As the touch panel **70***a*, there may be employed a touch panel of any suitable one of various types, including a resistance film type, a capacitance type, a surface acoustic

wave type, an infrared ray type, an electromagnetic induction type, an image recognition type, and an optical sensor type. Although the touch panel types include a type that detects a touch when a contact with the touch panel is detected and a type that detects a touch when proximity of a finger or a pen to the touch panel is detected, either type can be employed.

(51) Note that the digital camera **100** may be provided with an audio input section (not shown) that transmits audio signals obtained from a built-in microphone or an audio input device connected via an audio input terminal, to the system controller **50**. In this case, the system controller **50** selects input audio signals as required, converts the selected input audio signals from analog to digital, and performs processing for making the signal level appropriate, processing for reducing specific frequencies, and so forth.

(52) In the present embodiment, the user can set a method of designating a position index (e.g. the position of an AF frame) in a case where a touch-move operation is performed in an eye-contacted state to one of an absolute position designation method and a relative position designation method. The absolute position designation method refers to a method in which input coordinates on the touch panel **70***a* and display coordinates on the display surface of the EVF **29** are associated with each other. In the case of the absolute position designation method, when a touch-down on the touch panel **70***a* is detected, the AF frame is set to a position associated with the touched position (position where the coordinates thereof are input) (i.e. the AF frame is moved from a position before detecting the touch-down) even when no touch-move is detected. The position set by the absolute position designation method is a position based on the touched-down position regardless of the position set before the touch-down. Further, when a touch-move is detected after the touch-down, the position of the AF frame is also moved based on the touched position after the touch-move. The relative position designation method refers to a method in which input coordinates on the touch panel **70***a* and display coordinates on the display surface of the EVF **29** are not associated with each other. In the case of the relative position designation method, in a state in which only a touch-down on the touch panel **70***a* is detected, but a touch-move is not detected, the position of the AF frame is not moved from a position before the touch-down. When a touch-move is detected after that, the position of the AF frame is moved from the currently set position of the AF frame (position set before the touch-down) by a distance corresponding to the movement amount of the touch-move in a direction of movement of the touch-move regardless of the touched-down position.

(53) Note that the user can set one of a plurality of AF methods including “one point AF” and “entire area AF” as an AF area setting (AF frame-setting method). Further, the user can set whether or not to perform subject detection (tracking). The “one point AF” refers to a method in which a user designates one point as a position to perform AF using a one point AF frame. The “entire area AF” refers to a method in which in a case where a user does not designate a subject to be tracked, the AF position is automatically set based on an automatic selection condition. The tracking setting can be reflected on each of these AF area settings in a multiplied manner, and in a case where the tracking setting is set to be enabled, a mode is set in which a face is preferentially selected as an AF target subject if a face of a person is detected from an LV image. In a case where a plurality of faces are detected, one of the faces is selected to be set as the AF target subject according to priorities including a face having a large size, a face positioned close to the digital camera **100** (on the very near side), a face positioned close to the center within the screen, and a face of an individual registered in advance. If a face of a person is not detected, a subject other than a face is selected and set as the AF target subject according to priorities including a subject positioned close to the digital camera **100** (on the very near side), a subject having high contrast, a subject having high priority, such as an animal and a vehicle, and a moving body. In a case where a subject to be tracked is designated by a user, the subject to be tracked is set as the AF target subject. That is, the automatic selection condition is that by performing weighting using at least one of element conditions, enumerated below by way of example, an obtained score is not lower than a

predetermined threshold value or is the highest. The element conditions are as follows: a face of a detected person; a face having a large size; a face positioned close to the digital camera **100** (on the near side); a face positioned close to the center within the screen; a face of an individual registered in advance; a subject positioned close to the digital camera **100** (on the very near side); a subject having high contrast; a subject having high priority, such as an animal and a vehicle; and a moving body.
(54) FIG. **3** is a flowchart of a shooting mode process performed by the digital camera **100** shown in FIGS. **1**A and **1**B. The shooting mode process in FIG. **3** is realized by the system controller **50** that loads a program stored in the nonvolatile memory **56** into the system memory **52** and executes the loaded program. The shooting mode process in FIG. **3** is started when the digital camera **100** is started up in the shooting mode, and flags, control variables, and so forth are initialized.
(55) Referring to FIG. **3**, first, in a step S**301**, the system controller **50** displays a shooting live view **401** shown in FIG. **4**A on the display section **28** based on signals detected by the image capturing section **22**. Then, the shooting mode process proceeds to a step S**302**.
(56) In the step S**302**, the system controller **50** displays shooting-related information **402** appearing in FIG. **4**A on the shooting live view **401**. The shooting-related information **402** includes an icon indicating a set status of various settings and shooting parameters, such as a shutter speed, of the digital camera **100**. Further, on the shooting live view **401**, a GUI item for showing a result of analysis of a live view image to a user, such as a subject detection frame **403** appearing in FIG. **4**A, is displayed. When the step S**302** is completed, the shooting mode process proceeds to a step S**303**.
(57) In the step S**303**, the system controller **50** determines whether or not the user has performed an operation for switching the AF area setting, on the operation section **70**. For example, the user can perform an operation for switching the screen displayed on the display section **28** from the shooting live view **401** to a menu setting screen **500** shown in FIG. **5**, and perform an operation for switching the AF area setting from the menu setting screen **500**. On the menu setting screen **500**, not only the AF area setting, but also a tracking setting, a shutter method setting, a drive mode setting, and a shutter speed setting can be switched. If the user has performed the operation for switching the AF area setting, the shooting mode process proceeds to a step S**304**. If the user has not performed the operation for switching the AF area setting, the shooting mode process proceeds to a step S**305**.
(58) In the step S**304**, the system controller **50** changes the AF area setting according to the user's operation. More specifically, the system controller **50** changes the AF area setting to a setting selected by the user from “one point AF” and “entire area AF” displayed on the menu setting screen **500**. Then, the shooting mode process proceeds to the step S**305**.
(59) In the step S**305**, the system controller **50** determines whether or not the user has performed an operation for switching the tracking setting, on the operation section **70**. If the user has performed the operation for switching the tracking setting, the shooting mode process proceeds to a step S**306**. If the user has not performed the operation for switching the tracking setting, the shooting mode process proceeds to a step S**307**.
(60) In the step S**306**, the system controller **50** changes the tracking setting according to the user's operation. More specifically, the system controller **50** changes the tracking setting to a setting selected by the user from “set” and “not set” displayed on the menu setting screen **500**. Then, the shooting mode process proceeds to the step S**307**.
(61) In the step S**307**, the system controller **50** determines whether or not the user has performed an operation for switching the shutter method setting, on the operation section **70**. If the user has performed the operation for switching the shutter method setting, the shooting mode process proceeds to a step S**308**. If the user has not performed the operation for switching the shutter method setting, the shooting mode process proceeds to a step S**309**.
(62) In the step S**308**, the system controller **50** changes the shutter method setting according to the user's operation. More specifically, the system controller **50** changes the shutter method setting to a

setting selected by the user from “mechanical shutter” and “electronic shutter” displayed on the menu setting screen **500**. If the “mechanical shutter” is selected, the mechanical shutter method for controlling the shutter speed using a physical mechanism is set as the shutter method. If the “electronic shutter” is selected, the electronic shutter method for electronically controlling the image capturing device to adjust the shutter speed is set as the shutter method. Although in the present embodiment, the description is given of a case where the choices of the shutter method setting include the mechanical shutter method and the electronic shutter method, any other suitable method may be provided as a choice. For example, examples of the other choice of the shutter method include a method for performing electronically controlling only a front curtain (electronic front curtain). When the step S**308** is completed, the shooting mode process proceeds to the step S**309**.
(63) In the step S**309**, the system controller **50** determines whether or not the user has performed an operation for switching the drive mode setting, on the operation section **70**. If the user has performed the operation for switching the drive mode setting, the shooting mode process proceeds to a step S**310**. If the user has not performed the operation for switching the drive mode setting, the shooting mode process proceeds to a step S**311**.
(64) In the step S**310**, the system controller **50** changes the drive mode setting according to the user's operation. In the drive mode setting, a frame speed in continuous shooting is set. The frame speed is a parameter indicating the number of images shot per unit time in continuous shooting. The frame speed can also be referred to as the continuous shooting speed. In the step S**310**, more specifically, the system controller **50** changes the drive mode setting to a setting selected by the user from “high speed (30 frames/sec)”, “normal (10 frames/sec)”, and “low speed (3 frames/sec)” displayed on the menu setting screen **500**. Then, the shooting mode process proceeds to the step S**311**.
(65) In the step S**311**, the system controller **50** determines whether or not the user has performed an operation for switching the shutter speed setting, on the operation section **70**. If the user has performed the operation for switching the shutter speed setting, the shooting mode process proceeds to a step S**312**. If the user has not performed the operation for switching the shutter speed setting, the shooting mode process proceeds to a step S**313**.
(66) In the step S**312**, the system controller **50** changes the shutter speed setting according to the user's operation. More specifically, the system controller **50** changes the shutter speed setting to a setting selected by the user on the menu setting screen **500**. Then, the shooting mode process proceeds to the step S**313**.
(67) In the step S**313**, the system controller **50** performs a shooting process, described hereinafter with reference to FIG. **6**. Then, in a step S**314**, the system controller **50** determines whether or not the user has performed any other operation on the operation section **70**. The other operation refers to e.g. an operation for changing various parameters (such as an aperture value) associated with shooting. If the user has performed another operation, the shooting mode process proceeds to a step S**315**. If the user has performed no other operation, the shooting mode process proceeds to a step S**316**.
(68) In the step S**315**, the system controller **50** performs other processing according to the user's operation. The other processing refers to e.g. processing for changing various parameters (such as the aperture value) associated with shooting. Then, in the step S**316**, the system controller **50** determines whether or not the user has performed a termination operation, on the operation section **70**. If the user has not performed the termination operation, the shooting mode process returns to the step S**302**. If the user has performed the termination operation, the shooting mode process is terminated.
(69) Incidentally, in shooting using the electronic shutter, there is no shutter sound generated by driving the mechanical shutter, and hence the user cannot always recognize the shooting timing. To cope with this problem, the digital camera **100** lights a GUI item for notifying a user of the

shooting timing, such as a white frame item **405**, described hereinafter with reference to FIG. **4**C, on the display section **28** during an exposure processing execution period which is a time period over which exposure processing is performed. The user can recognize the timing of continuous shooting based on the display state of the white frame item **405** even when there is no shutter sound.

(70) However, there is a case where even when the white frame item **405** is lighted on the display section **28**, the user cannot recognize the shooting timing. For example, in a case where the shutter speed is high and the frame speed is low, the exposure processing execution period is reduced, and hence even when the white frame item **405** is lighted on during the exposure processing execution period, the lighting on time of the white frame item **405** is too short, so that there is a possibility that the user is incapable of recognizing the shooting timing. Further, in a case where the frame speed is high, an interval between one exposure processing execution period and the next exposure processing execution period is short, and hence there is a possibility that the user is incapable of recognizing a pause in the lighting of the white frame item **405**.

(71) To prevent this, in the present embodiment, a minimum lighting-on period over which a lighted on state (displayed state) of the white frame item **405** is continued is set. Here, the minimum lighting-on period is assumed to be 16 msec. In a case where continuous shooting is performed at a certain frame speed and a second shutter speed higher than a first shutter speed, the digital camera **100** continues to light on (display) the white frame item **405** for the minimum lighting-on period longer than an exposure processing executing period determined based on the second shutter speed.

(72) Further, in the present embodiment, a minimum lighting-off period over which a lighted off state (non-displayed state) of the white frame item **405** is continued is set. Here, the minimum lighting-off period is assumed to be 30 msec. The minimum lighting-off period is longer than the minimum lighting-on period. In a case where continuous shooting is performed at a certain shutter speed and a second frame speed higher than a first frame speed, when the minimum lighting-off period elapses which is longer than a time period from an end of lighting on (displaying) of the white frame item **405** to a start of execution of the next exposure processing, which is determined based on the second frame speed, the digital camera **100** lights on again (displays again) the white frame item **405**.

(73) FIGS. **6**A and **6**B are a flowchart of the shooting process performed in the step S**313** in FIG. **3**.

(74) Referring to FIG. **6**A, in a step S**601**, the system controller **50** determines whether or not a user has operated the first shutter switch **62**. If the user has operated the first shutter switch **62**, the shooting process proceeds to a step S**602**. If the user has not operated the first shutter switch **62**, the shooting process is terminated.

(75) In the step S**602**, the system controller **50** performs shooting preparation operations, such as photometry and ranging. FIG. **4**B shows an example of a screen displayed on the display section **28** when the shooting preparation operations are performed. On this screen, the shooting-related information **402** narrowed down to information necessary for the shooting preparation operations is displayed in a state superimposed on the shooting live view **401**. Further, on this screen, not the subject detection frame **403**, but an AF frame **404** indicating a photometry result is displayed in a state superimposed on the shooting live view **401**. Then, the shooting process proceeds to a step S**603**.

(76) In the step S**603**, the system controller **50** determines whether or not the user has operated the second shutter switch **64**. If the user has not operated the second shutter switch **64**, the shooting process returns to the step S**601**. If the user has operated the second shutter switch **64**, the shooting process proceeds to a step S**604**.

(77) In the step S**604**, the system controller **50** starts shooting. The following description will be given of a case where continuous shooting is performed. Then, in a step S**605**, the system controller **50** determines whether or not the operation of the second shutter switch **64** is being continued. If

the operation of the second shutter switch **64** is being continued, the shooting process proceeds to a step S**607**. If the operation of the second shutter switch **64** is not being continued, the shooting process proceeds to a step S**606**.
(78) In the step S**606**, the system controller **50** terminates the shooting (continuous shooting). Then, the shooting process returns to the step S**603**. At this time, a lighting-on flag and a lighting-off flag, described hereinafter, are all initialized.
(79) In the step S**607**, the system controller **50** determines whether or not to start the exposure processing based on a shutter speed set on the menu setting screen **500**. If it is determined in the step S**607** that the exposure processing is not to be started, i.e. the exposure processing is being executed, the shooting process proceeds to a step S**614** in FIG. **6**B, described hereinafter. If it is determined in the step S**607** that the exposure processing is to be started, the shooting process proceeds to a step S**608**.
(80) In the step S**608**, the system controller **50** instructs the start of the exposure processing. With this, in the digital camera **100**, the exposure processing is started. Then, the shooting process proceeds to a step S**609**. In the step S**609**, the system controller **50** determines whether or not the white frame item **405**, described hereinafter, is being lighted on the shooting live view **401**.
(81) If it is determined in the step S**609** that the white frame item **405** is being lighted on the shooting live view **401**, the shooting process proceeds to the step S**614** in FIG. **6**B. If it is determined in the step S**609** that the white frame item **405** is not being lighted on the shooting live view **401**, the shooting process proceeds to a step S**610**. In the step S**610**, the system controller **50** determines whether or not it is during the minimum lighting-off period of the white frame item **405**.
(82) If it is determined in the step S**610** that it is not during the minimum lighting-off period of the white frame item **405**, the shooting process proceeds to a step S**611**.
(83) In the step S**611**, the system controller **50** displays the live view image and the white frame item **405** together on the shooting live view **401** as shown in FIG. **4**C. FIG. **4**C shows an example of a screen displayed on the display section **28** when the exposure processing is executed in continuous shooting. On this screen, the shooting-related information **402** narrowed down to the information necessary for shooting, the AF frame **404** indicating a photometry result, and the white frame item **405** indicating that the exposure processing is being executed are displayed in a state superimposed on the shooting live view **401**.
(84) The white frame item **405** is displayed such that the white frame item **405** surrounds an outer periphery of the live view screen. This enables a user to easily confirm a state of a subject displayed on the shooting live view **401**. Further, it is possible to obtain an effect of giving a warning against handshake to a user while the white frame item **405** is being displayed. Further, since the white color, which is bright and noticeable, is employed for the frame color of the white frame item **405**, it is possible to enable the user to easily recognize the white frame item **405** at a peripheral end of sight of the user even though the white frame item **405** is at the outer periphery of the live view screen.
(85) Note that the form of the white frame item **405** is not limited to this, but for example, another noticeable color different from white may be employed for the frame color. Further, the form of the white frame item **405** may be dynamically changed according to a state of the camera. For example, in a case where a user's line of sight is directed toward the vicinity of the outer periphery of the screen, the transmittance of the white frame item **405** in the vicinity of the outer periphery of the screen may be controlled to be reduced to thereby enable the user to more easily confirm a state of a subject displayed on the shooting live view **401**. Then, in a step S**612**, the system controller **50** starts measurement of a lighting-on timer for determining expiration of the minimum lighting-on period of the white frame item **405**, and the shooting process proceeds to the step S**614**.
(86) If it is determined in the step S**610** that it is during the minimum lighting-off period of the white frame item **405**, the shooting process proceeds to a step S**613**. In the step S**613**, the system

controller **50** sets the lighting-on flag for controlling causing the white frame item **405** to be lighted on in a state in which the white frame item **405** remains lighted off on the shooting live view **401**. That is, even when the exposure processing is started, if it is during the minimum lighting-off period of the white frame item **405**, the lighted off state of the white frame item **405** is continued. In other words, even when the exposure processing for the next shooting is started before the minimum lighting-off period elapses after completion of display of the item associated with the preceding shooting, the system controller **50** controls the display not to start the display of the item. Then, the shooting process proceeds to the step S**614** in FIG. **6**B.
(87) In the step S**614**, the system controller **50** determines, based on the shutter speed set on the menu setting screen **500**, whether or not to terminate the exposure processing.
(88) If it is determined in the step S**614** that the exposure processing is not to be terminated, the shooting process proceeds to a step S**623**. If it is determined in the step S**614** that the exposure processing is to be terminated, the shooting process proceeds to a step S**615**. In the step S**615**, the system controller **50** instructs termination of the exposure processing. The digital camera **100** terminates the exposure processing in response to this instruction. Then, in a step S**616**, the system controller **50** starts execution of predetermined processing. The predetermined processing includes processing performed in a time period from an end of one exposure processing operation to a start of the next exposure processing operation, such as processing for developing an image read from the sensor and processing for writing the image subjected to the developing processing into a recording medium. Then, the shooting process proceeds to a step S**617**. In the step S**617**, the system controller **50** determines whether or not the white frame item **405** is lighted off on the shooting live view **401**.
(89) If it is determined in the step S**617** that the white frame item **405** is lighted off, the shooting process proceeds to a step S**618**. In the step S**618**, the system controller **50** sets the lighting-on flag to off. Then, the shooting process proceeds to the step S**623**.
(90) If it is determined in the step S**617** that the white frame item **405** is not lighted off, the shooting process proceeds to a step S**619**. In the step S**619**, the system controller **50** determines whether or not it is during the minimum lighting-on period of the white frame item **405**.
(91) If it is determined in the step S**619** that it is not during the minimum lighting-on period of the white frame item **405**, the shooting process proceeds to a step S**620**. In the step S**620**, the system controller **50** causes the white frame item **405** to be lighted on the shooting live view **401** as shown in FIG. **4**D. In a case where the display of the white frame item **405** is started when the exposure processing is started, a time period from the start of exposure to the end of exposure is the display period of the white frame item **405**.
(92) FIG. **4**D shows an example of a screen displayed on the display section **28** after one exposure processing operation is terminated until the next exposure processing operation is started in the continuous shooting. On this screen, the shooting-related information **402** narrowed down to the information necessary for shooting, and the AF frame **404** indicating a photometry result are displayed in a state superimposed on the shooting live view **401**, but the white frame item **405** is not displayed. Then, the shooting process proceeds to a step S**621**. In the step S**621**, the system controller **50** starts measurement of a lighting-off timer for determining expiration of the minimum lighting-off period of the white frame item **405**. Also, the system controller **50** stops measurement of the lighting-on timer. Then, the shooting process proceeds to the step S**623**.
(93) If it is determined in the step S**619** that it is during the minimum lighting-on period of the white frame item **405**, the shooting process proceeds to a step S**622**. In the step S**622**, the system controller **50** sets the lighting-off flag for controlling causing the white frame item **405** to be lighted off to on while causing the white frame item **405** to remain lighted on the shooting live view **401**. That is, in the present embodiment, even when the exposure processing is terminated, the lighted on state of the white frame item **405** is continued if it is during the minimum lighting-on period of the white frame item **405**. In other words, in a case where the exposure time from the start to the

end of exposure is shorter than the minimum lighting-on period, the system controller **50** continues to display the white frame item even after the exposure time has elapsed. That is, the system controller **50** continues to display the white frame item during the minimum lighting-on period regardless of the exposure time.

(94) Then, the shooting process proceeds to the step S**623**. In the step S**623**, the system controller **50** determines, based on a result of the measurement performed by the lighting-on timer, whether or not the minimum lighting-on period has expired.

(95) If it is determined in the step S**623** that the minimum lighting-on period has expired, the shooting process proceeds to a step S**624**. In the step S**624**, the system controller **50** determines whether or not the lighting-off flag is on.

(96) If it is determined in the step S**624** that the lighting-off flag is on, the shooting process proceeds to a step S**625**. In the step S**625**, the system controller **50** sets the lighting-off flag to off. Then, in a step S**626**, the system controller **50** causes the white frame item **405** to be lighted off. Then, in a step S**627**, the system controller **50** starts measurement of the lighting-off timer. Also, the system controller **50** stops measurement of the lighting-on timer, if the system controller **50** makes measurement of the lighting-on timer. Then, the shooting process proceeds to a step S**628**.

(97) Also, it is determined in the step S**623** that the minimum lighting-on period has not expired or if it is determined in the step S**624** that the lighting-off flag is not on, the shooting process proceeds to the step S**628**. In the step S**628**, the system controller **50** determines, based on a result of the measurement performed by the lighting-off timer, whether or not the minimum lighting-off period has expired.

(98) If it is determined in the step S**628** that the minimum lighting-off period has expired, the shooting process proceeds to a step S**629**. In the step S**629**, the system controller **50** determines whether or not the lighting flag is on.

(99) If it is determined in the step S**629** that the lighting-on flag is on, the shooting process proceeds to a step S**630**. In the step S**630**, the system controller **50** sets the lighting-on flag to off. Then, in a step S**631**, the system controller **50** lights on the white frame item **405**. Then, in a step S**632**, the system controller **50** starts measurement of the lighting-on timer and stops measurement of the lighting-off timer. Then, the shooting process returns to the step S**605**.

(100) If it is determined in the step S**628** that the minimum lighting-off period has not expired, or if it is determined in the step S**629** that the lighting-on flag is not set to on, the shooting process returns to the step S**605** in FIG. **6**A.

(101) Note that although in the above-described shooting process in FIGS. **6**A and **6**B, if it is determined in the step S**605** that the operation of the second shutter switch **64** has been terminated in continuous shooting, the shooting termination processing in the step S**606** is immediately executed, but this is not limitative. For example, the shooting termination processing may be executed after completion of processing being executed, or the shooting process may be continued until the minimum lighting-on period or the minimum lighting-off period of the white frame item **405** expires.

(102) Further, in the above-described shooting process in FIGS. **6**A and **6**B, the minimum lighting-off period of the white frame item **405** is longer than the minimum lighting-on period. In order for a user to recognize that a bright color, such as white, is lighted off, it is necessary to continue the lighted off state for a relatively long time period. By setting the minimum lighting-off period of the white frame item **405** to be longer than the minimum lighting-on period of the same, it is possible to realize natural blinking which the user can recognize.

(103) FIGS. **7**A to **7**E are diagrams useful in explaining states of the white frame item **405** lighted on and lighted off by the shooting process in FIGS. **6**A and **6**B for each combination of a shutter speed and a frame speed.

(104) FIG. **7**A shows an explanatory note used for explaining FIGS. **7**B to **7**E. Referring to FIG. **7**A, reference numeral **701** denotes a time period within the minimum lighting-on period, during

which the white frame item **405** is lighted on. Reference numeral **702** denotes a time period other than the minimum lighting-on period, during which the white frame item **405** is lighted on. Reference numeral **703** denotes a time period within the minimum lighting-off period, during which the white frame item **405** is lighted off. Reference numeral **704** denotes a time period other than the minimum lighting-off period, during which the white frame item **405** is lighted off.

(105) FIGS. **7**B to **7**E show the lighted on and lighted off states of the white frame item **405** associated with changes in a waveform **712** representing the exposure processing execution period based on parameter settings **711** including the shutter speed setting and the frame speed setting. In the waveform **712**, a period of 1 indicates the exposure processing execution period, and a period of 0 indicates a period from completion of one exposure processing operation to the start of the next exposure processing operation (hereinafter referred to as the "exposure processing non-execution period"). In the period of 0, the exposure processing is not executed, but processing other than the exposure processing is executed.

(106) FIG. **7**B shows an example of a case where the shutter speed is relatively low and the frame speed is relatively low. In the digital camera **100**, as the shutter speed is lower, the exposure processing execution period is longer. For this reason, in a case where the shutter speed is relatively low, as shown in FIG. **7**B, the white frame item **405** is lighted on during each period of 1 in the waveform **712** i.e. during the exposure processing execution period, whereby the user can easily recognize the lighted on state of the white frame item **405**. On the other hand, in the digital camera **100**, as the frame speed is lower, time to elapse until the start of execution of the next exposure processing is longer. For this reason, in a case where the frame speed is relatively low, as shown in FIG. **7**B, the white frame item **405** is lighted off during each period of 0 in the waveform **712**, i.e. during the exposure processing non-execution period, whereby the user can easily recognize the lighted off state of the white frame item **405**.

(107) FIG. **7**C shows an example of a case where the shutter speed is relatively low and the frame speed is relatively high. In the digital camera **100**, in a case where the shutter speed is relatively low, as shown in FIG. **7**B described above, the white frame item **405** is lighted on during the exposure processing execution period, whereby the user can easily recognize the lighted on state of the white frame item **405**. On the other hand, in the digital camera **100**, as the frame speed is higher, time to elapse until the start of execution of the next exposure processing is shorter. For this reason, if the white frame item **405** is only lighted off during the exposure processing non-execution period as described above with reference to FIG. **7**B, an interval between one exposure processing execution period and the next exposure processing execution period is so short that it is difficult for the user to recognize a pause in the lighting of the white frame item **405**. To cope with this, in the present embodiment, a minimum lighting-off period **721** is provided, and the lighted off state of the white frame item **405** is continued at least during the minimum lighting-off period **721**. This makes it possible to sufficiently secure the lighting off time of the white frame item **405** indicating the timing of continuous shooting, and enable the user to easily recognize a pause in the lighting of the white frame item **405**. As a result, the user can easily recognize the timing of continuous shooting.

(108) FIG. **7**D shows an example of a case where the shutter speed is relatively high and the frame speed is relatively low. In the digital camera **100**, as the shutter speed is higher, the exposure processing execution period is shorter. For this reason, if the white frame item **405** is only lighted on during the exposure processing execution period as shown in FIGS. **7**B and **7**C, the lighting on time of the white frame item **405** is so short that it is difficult for the user to recognize the lighting of the white frame item **405**. To cope with this, in the present embodiment, a minimum lighting-on period **722** is provided, and the lighted on state of the white frame item **405** is continued at least during the minimum lighting-on period **722**. This makes it possible to sufficiently secure the lighting on time of the white frame item **405** indicating the timing of continuous shooting, and enable the user to easily recognize the lighting of the white frame item **405**. As a result, the user

can easily recognize the timing of continuous shooting. On the other hand, in a case where the frame speed is relatively low, by lighting off the white frame item **405** during the exposure processing non-execution period as described above with reference to e.g. FIG. **7**B, the user can easily recognize the lighting off of the white frame item **405**.

(109) FIG. **7**E shows an example of a case where the shutter speed is relatively high and the frame speed is also relatively high. In this example, as described above, the lighted on state of the white frame item **405** is continued at least during a minimum lighting-on period **723**, and hence it is possible to enable the user to easily recognize the lighting of the white frame item **405**. Further, as described above, the lighted off state of the white frame item **405** is continued at least during a minimum lighting-off period **724**. With this, it is possible to enable the user to easily recognize a pause in the lighting of the white frame item **405**. Note that in FIG. **7**E, the number of times of blinking of the white frame item **405** does not necessarily coincide with the number of actual shooting times (number of frames). Here, in the case where the shutter speed is relatively high and the frame speed is also relatively high, even when the white frame item **405** is lighted on and lighted off the number of times corresponding to the number of actual shooting times, blinking of the white frame item **405** is so fast that it is difficult for the user to grasp the number of actual shooting times from the number of blinking times of the white frame item **405**. In this case, it is enough for the user to be notified of execution of high-speed continuous shooting rather than to be notified of the exact number of shooting times. On the other hand, in the present embodiment, blinking (lighting on and lighting off) of the white frame item **405** is performed at intervals roughly equivalent to those of high-speed continuous shooting, based on the minimum lighting-on period and the minimum lighting-off period. With this, it is possible to notify the user that high-speed continuous shooting is being performed in a case where the shutter speed is relatively high and the frame speed is also relatively high.

(110) The present invention has been described heretofore based on the above-described embodiment, but the present invention is not limited to the above-described embodiment. For example, a mechanism for generating an electronic shutter sound may be provided to notify a user of the shooting timing using the electronic shutter sound. Note that the electronic shutter sound is generated strictly in synchronism with the shooting timing, but blinking of the white frame item **405** is not necessarily required to coincide with the shooting timing as shown in FIG. **7**E.

(111) Further, in the above-described embodiment, the display of the white frame item **405** may be performed only in shooting using an electronic shutter. Here, when shooting using a mechanical shutter is performed, the shooting timing and the exposure processing execution period can be notified by a shutter sound of the mechanical shutter, and hence the display of the white frame item **405** surrounding the shooting live view **401** is not required. Therefore, by limiting the display of the white frame item **405** to the case of shooting using an electronic shutter, it is possible, at the time of shooting using a mechanical shutter, to enable a user to easily confirm the shooting live view **401** during shooting using the mechanical shutter.

(112) Although in the above-described embodiment, the white frame item **405** is lighted on during the exposure processing execution period, and lighted off during the exposure processing non-execution period, this is not limitative. For example, the white frame item **405** may be lighted off during the exposure processing execution period, and lighted on during the exposure processing non-execution period.

(113) In the above-described embodiment, the GUI item for notifying a user of the shooting timing is not limited to a white frame displayed on the outer periphery of the screen, such as the white frame item **405**. For example, as the GUI item for notifying a user of the shooting timing, a frame **406** of an unnoticeable color, such as gray, may be displayed in the vicinity of the center of the screen, as shown in FIG. **4**E.

(114) Further, in the above-described embodiment, as the GUI item for notifying a user of the shooting timing, an icon **407** appearing in FIG. **4**F may be displayed in a blinking state.

(115) Although in the above-described embodiment, the configuration of displaying the white frame item **405** on the display section **28** is described, this is not limitative. For example, the white frame item **405** may be displayed on one of the display section **28** and the EVF **29**, as a unit whose display is set to on i.e. on which the shooting live view **401** is displayed.
(116) Further, in the above-described embodiment, the display form of the white frame item **405** may be varied between the display section **28** and the EVF **29**.
(117) Although in the above-described embodiment, one shooting period is divided into the exposure processing execution period and the exposure processing non-execution period, this is not limitative. For example, when continuous shooting is performed in a mode (SERVO AF) in which the focus is automatically and continuously adjusted, a GUI item for notifying a user of the shooting timing may be lighted on during the SERVO driving and lighted off in a time period in which focusing is achieved and the SERVO driving is temporarily stopped.
(118) In the above-described embodiment, the display of the white frame item **405** may be performed such that, in continuous shooting, the white frame item **405** is not displayed at a first shooting timing and is displayed at second and subsequent shooting timings.
(119) Note that the above-described various controls described as performed by the CPU may be performed by one hardware item or by a plurality of hardware items that share the processing operations, for control of the whole apparatus.
OTHER EMBODIMENTS
(120) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(121) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(122) This application claims the benefit of Japanese Patent Application No. 2020-209347, filed Dec. 17, 2020, which is hereby incorporated by reference herein in its entirety.

Claims

1. An electronic apparatus, comprising: a memory and at least one processor which function as: a first setting unit configured to set a shutter speed; a second setting unit configured to set a continuous shooting speed; a third setting unit configured to set a shutter method from methods at least including a mechanical shutter method and an electronic shutter method; a control unit

configured to, during continuous shooting, control so as to display, on a display, an item indicating a timing of execution of shooting together with an image, for a display time period associated with an exposure time which is set based on the shutter speed set by the first setting unit, and so as not to display the item for a non-display time period from an end of the display time period until a start of next display time period, wherein the control unit, in a case where a length of the exposure time is shorter than a predetermined threshold value, controls so that a display time period of the item is a first length which is independent from the exposure time, wherein the control unit, in a case where a high continuous shooting speed is set by the second setting unit, controls so as to display the item for a number of times less than a number of shooting times in a continuous shooting, and wherein in a case where the electronic shutter method is set for the shutter method by the third setting unit when images captured by an image capturing device are continuously displayed on the display as a live view image during the continuous shooting, the control unit controls so as to display the item on the display together with the live view image, and in a case where the mechanical shutter method is set for the shutter method by the third setting unit when the live view image is displayed on the display during the continuous shooting, the control unit controls so as to display the live view image on the display without displaying the item on the display.

2. The electronic apparatus according to claim 1, wherein continuous shooting using the mechanical shutter method for controlling the shutter speed using a physical mechanism or continuous shooting using the electronic shutter method for electronically controlling the image capturing device to control the shutter speed, can be performed in the continuous shooting.

3. The electronic apparatus according to claim 2, wherein the control unit controls, during the continuous shooting using the electronic shutter method, so as to display the item indicating the timing of execution of shooting.

4. The electronic apparatus according to claim 3, wherein the control unit controls so as to display the item only during the continuous shooting using the electronic shutter method, and controls so as not to display the item during the continuous shooting using the mechanical shutter method.

5. The electronic apparatus according to claim 2, further comprising a continuous shooting setting unit configured to set whether to execute the continuous shooting using the mechanical shutter method or to execute the continuous shooting using the electronic shutter method.

6. The electronic apparatus according to claim 1, wherein the control unit controls the display of the item so that the length of the display time period is the first length or more and that the length of the non-display time period is a second length or more.

7. The electronic apparatus according to claim 6, wherein the control unit controls, in a case where the high continuous shooting speed is set by the second first setting unit, to display the item using the display time period based on the first length and the non-display time period based on the second length.

8. The electronic apparatus according to claim 6, wherein the second length is longer than the first length.

9. The electronic apparatus according to claim 8, wherein the item is a white item.

10. The electronic apparatus according to claim 1, wherein the control unit controls to display the item on an edge area of the display.

11. The electronic apparatus according to claim 1, wherein the item is a frame that surrounds an outer periphery of the live view image.

12. The electronic apparatus according to claim 1, wherein the predetermined threshold value is the first length.

13. The electronic apparatus according to claim 1, wherein the continuous shooting speed is a frame speed, and the high continuous shooting speed is larger than 10 frames/sec.

14. The electronic apparatus according to claim 1, wherein the continuous shooting speed is a frame speed, and the high continuous shooting speed is equal to or larger than 30 frames/sec.

15. The electronic apparatus according to claim 1, wherein the control unit, in a case where the

high continuous shooting speed is set by the second setting unit and a high shutter speed is set by the first setting unit, controls so as to display the item for a number of times less than a number of shooting times in a continuous shooting.

16. The electronic apparatus according to claim 15, wherein the high shutter speed is equal to or higher than a shutter speed of 1/1000 seconds.

17. A method of controlling an electronic apparatus comprising: a first setting step of setting a shutter speed; a second setting step of setting a continuous shooting speed; a third setting step of setting a shutter method from methods at least including a mechanical shutter method and an electronic shutter method; during continuous shooting, controlling so as to display, on a display, an item indicating a timing of execution of shooting together with an image, for a display time period associated with an exposure time which is set based on the shutter speed set in the first setting step, and so as not to display the item for a non-display time period from an end of the display time period until a start of next display time period, wherein the controlling further comprises: in a case where a length of the exposure time is shorter than a predetermined threshold value, controlling so that a display time period of the item is a first length which is independent from the exposure time, in a case where a high continuous shooting speed is set by the second setting step, controlling so as to display the item for a number of times less than a number of shooting times in a continuous shooting, in a case where the electronic shutter method is set for the shutter method by the third setting step when images captured by an image capturing device are continuously displayed on the display as a live view image during the continuous shooting, controlling so as to display the item on the display together with the live view image, and in a case where the mechanical shutter method is set for the shutter method by the third setting step when the live view image is displayed on the display during the continuous shooting, controlling so as to display the live view image on the display without displaying the item on the display.

18. A non-transitory computer-readable storage medium storing a program that, when executed by a processor, cause the processor to perform operations comprising: a first setting step of setting a shutter speed; a second setting step of setting a continuous shooting speed; a third setting step of setting a shutter method from methods at least including a mechanical shutter method and an electronic shutter method; during continuous shooting, controlling so as to display, on a display, an item indicating a timing of execution of shooting together with an image, for a display time period associated with an exposure time which is set based on the shutter speed set in the first setting step, and so as not to display the item for a non-display time period from an end of the display time period until a start of next display time period, wherein the controlling further comprises: in a case where a length of the exposure time is shorter than a predetermined threshold value, controlling so that a display time period of the item is a first length which is independent from the exposure time, in a case where a high continuous shooting speed is set by the second setting step, controlling so as to display the item for a number of times less than a number of shooting times in a continuous shooting, in a case where the electronic shutter method is set for the shutter method by the third setting step when images captured by an image capturing device are continuously displayed on the display as a live view image during the continuous shooting, controlling so as to display the item on the display together with the live view image, and in a case where the mechanical shutter method is set for the shutter method by the third setting step when the live view image is displayed on the display during the continuous shooting, controlling so as to display the live view image on the display without displaying the item on the display.